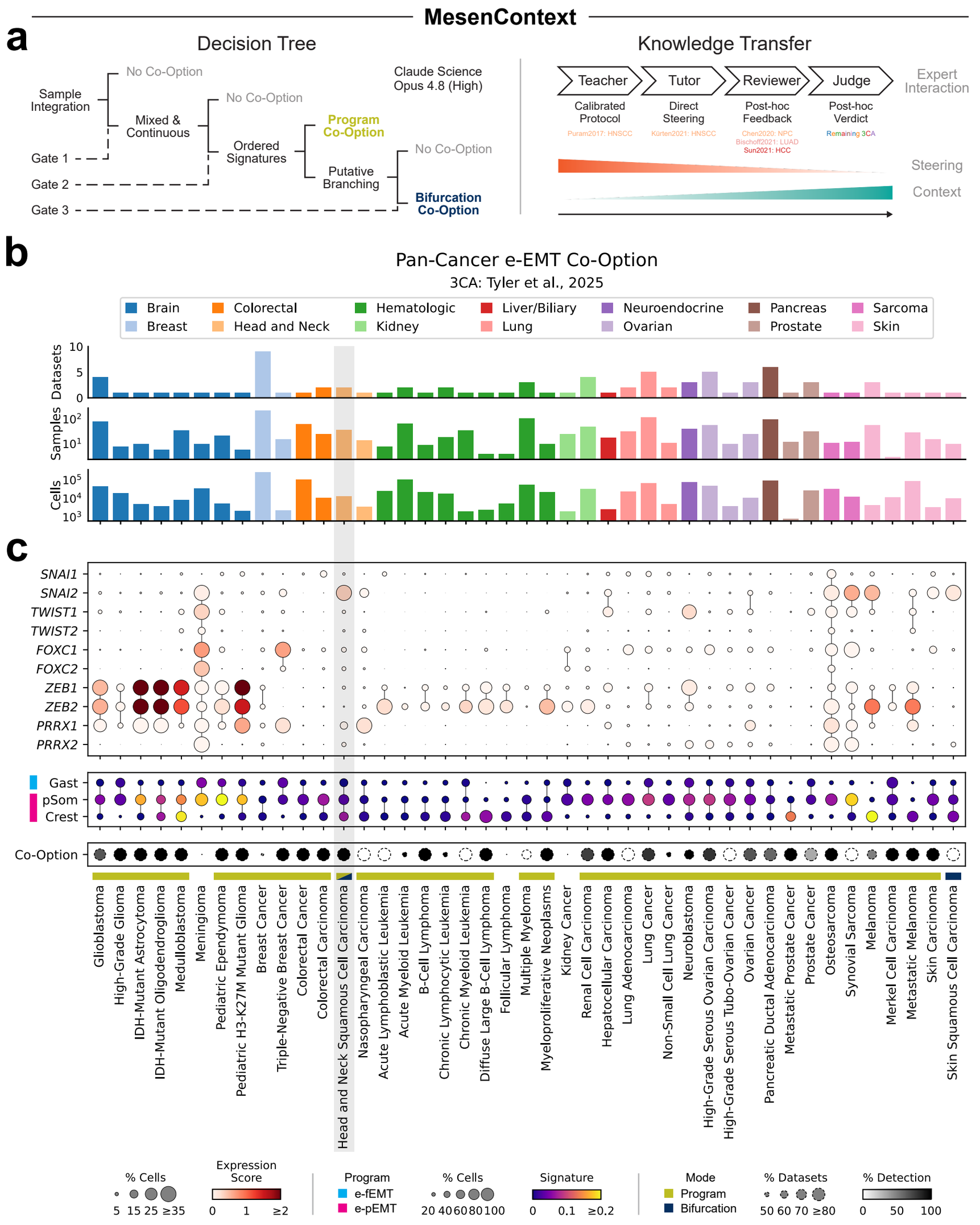

MesenContext
a
Decision Tree
Sample Integration
No Co-Option
Mixed & Continuous
No Co-Option
Ordered Signatures
Program Co-Option
Putative Branching
No Co-Option
Bifurcation Co-Option
Gate 1
Gate 2
Gate 3
Claude Science Opus 4.8 (High)
Knowledge Transfer
Teacher
Tutor
Reviewer
Judge
Expert Interaction
Calibrated Protocol
Direct Steering
Post-hoc Feedback
Post-hoc Verdict
Puram2017: HNSCC
Kürten2021: HNSCC
Chen2020: NPC
Bischoff2021: LUAD
Sun2021: HCC
Remaining 3CA
Steering
Context
b
Pan-Cancer e-EMT Co-Option
3CA: Tyler et al., 2025
Brain
Breast
Colorectal
Head and Neck
Hematologic
Kidney
Liver/Biliary
Lung
Neuroendocrine
Ovarian
Pancreas
Prostate
Sarcoma
Skin
Datasets
Samples
Cells
c
SNAI1
SNAI2
TWIST1
TWIST2
FOXC1
FOXC2
ZEB1
ZEB2
PRRX1
PRRX2
Gast
pSom
Crest
Co-Option
Glioblastoma
High-Grade Glioma
IDH-Mutant Astrocytoma
IDH-Mutant Oligodendroglioma
Medulloblastoma
Meningioma
Pediatric Ependymoma
Pediatric H3-K27M Mutant Glioma
Breast Cancer
Triple-Negative Breast Cancer
Colorectal Cancer
Colorectal Carcinoma
Head and Neck Squamous Cell Carcinoma
Nasopharyngeal Carcinoma
Acute Lymphoblastic Leukemia
Acute Myeloid Leukemia
B-Cell Lymphoma
Chronic Lymphocytic Leukemia
Chronic Myeloid Leukemia
Diffuse Large B-Cell Lymphoma
Follicular Lymphoma
Multiple Myeloma
Myeloproliferative Neoplasms
Kidney Cancer
Renal Cell Carcinoma
Hepatocellular Carcinoma
Lung Adenocarcinoma
Lung Cancer
Non-Small Cell Lung Cancer
Neuroblastoma
High-Grade Serous Ovarian Carcinoma
High-Grade Serous Tubo-Ovarian Cancer
Ovarian Cancer
Pancreatic Ductal Adenocarcinoma
Metastatic Prostate Cancer
Prostate Cancer
Osteosarcoma
Synovial Sarcoma
Melanoma
Merkel Cell Carcinoma
Metastatic Melanoma
Skin Carcinoma
Skin Squamous Cell Carcinoma
% Cells
Expression Score
Program
e-fEMT
e-pEMT
% Cells
Signature
Mode
Program
Bifurcation
% Datasets
% Detection